Skills Summary

Transformations of Absolute-Value Functions

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading the Vertex and the Transformations

Vertex form: $g(x) = a\,|x-h| + k$. **Vertex** (h, k); **axis of symmetry** $x = h$.

What each letter does to $y = |x|$**:** h shifts horizontally (right when $x-h$, left when $x+h$); k shifts vertically; $a < 0$ flips the V so it opens down; $|a| > 1$ makes it narrower, $0 < |a| < 1$ makes it wider.

Fast way to find h**:** set the inside of the bars equal to 0 and solve.

Example: Find the vertex of $g(x) = -2\,|x+5| + 3$ and describe the graph.
Step 1. The vertex is the one point where the distance inside the bars is zero, so set the inside to zero and read k off the outside.
Step 2. $x + 5 = 0 \Rightarrow x = -5$; then $g(-5) = -2\,|-5+5| + 3 = 3$. Since $a = -2$, the V opens down and is narrower than $y = |x|$.
$\Rightarrow$ vertex **$(-5,\ 3)$**

Try it: Find the vertex of $g(x) = 4\,|x-6| - 1$.

check: **$(6, -1)$**

(!) Watch out: $|x+3|$ shifts LEFT, not right — rewrite it as $|x-(-3)|$ and the sign stops fooling you. And a multiplies only the bars, so in $3\,|x+2| - 7$ the 3 never touches the -7.

Skill 2 — Sketching the Graph from Vertex Form

Three steps: (1) plot the vertex (h, k); (2) from the vertex go right 1 and up a to get a point on the right arm, then left 1 and up a for the left arm; (3) draw two straight rays — an absolute-value graph is never curved.

x-intercepts: solve $a\,|x-h| + k = 0$. They are always the same distance left and right of $x = h$.

Example: Sketch $y = |x-1| - 4$ and give its x-intercepts.
Step 1. Vertex form is already given, so plot the vertex first and then find where the two arms cross the axis.
Step 2. Vertex: $x - 1 = 0 \Rightarrow x = 1$, and $y = -4$ there. Intercepts: $|x-1| = 4 \Rightarrow x - 1 = \pm 4$.
$\Rightarrow$ vertex **$(1,\ -4)$**, $x =$ **-3** or **5**

Try it: Give the vertex and both x-intercepts of $y = |x+2| - 3$.

check: **$(-2, -3)$**, $x =$ **-5** or **1**

Skill 3 — Solving with the Graph in Mind

Why there are two answers: a horizontal line above the vertex of an upward V cuts it twice, and the two crossings sit the same distance on either side of $x = h$.

Method: isolate the bars, then split. $a\,|x-h| + k = c \;\Rightarrow\; |x-h| = \dfrac{c-k}{a} \;\Rightarrow\; x = h \pm \dfrac{c-k}{a}$.

Example: Solve $3\,|x-2| - 5 = 7$.
Step 1. The bars are not alone yet, so isolate them first; only then can the V be split into its two arms.
Step 2. $3\,|x-2| = 12 \Rightarrow |x-2| = 4 \Rightarrow x = 2 \pm 4$.
$\Rightarrow \quad x = \mathbf{-2}$ or $x = \mathbf{6}$

Try it: Solve $2\,|x+1| - 3 = 9$.

check: $x = \mathbf{-7}$ or $x = \mathbf{5}$

(!) Watch out: divide by a BEFORE splitting the bars. Splitting $3\,|x-2| = 12$ into $3x - 2 = \pm 12$ loses the bars and both answers.

Skill 4 — Writing the Equation from the Graph

Read the vertex, then one more point. The vertex hands you h and k at a glance. Substitute any second point (x_1, y_1) into $y = a\,|x-h| + k$ and solve the resulting one-variable equation for a: $a = \dfrac{y_1 - k}{|x_1 - h|}$.

Sanity check: if the second point is BELOW the vertex, a must come out negative.

Example: A V has vertex $(2, 1)$ and passes through $(5, 7)$. Write its equation.
Step 1. The vertex gives two of the three unknowns for free, so the extra point is only needed to pin down the steepness a.
Step 2. $h = 2$, $k = 1$, so $7 = a\,|5-2| + 1 \Rightarrow 3a + 1 = 7 \Rightarrow a = 2$.
$\Rightarrow \quad y = \mathbf{2}\,|x-2| + \mathbf{1}$

Try it: A V has vertex $(-4, 6)$ and passes through $(-1, 0)$. Write its equation.

check: $y = \mathbf{-2}\,|x+4| + \mathbf{6}$